PROJECT PROPOSAL

National IMEI-Based Device Tracking and Management System

(Central Equipment Identity Register — CEIR — with Distributed Equipment Identity Register — EIR — Enforcement)

Document Control

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Foreword

This proposal has been assembled over several months of internal technical review, consultation with regional regulatory counterparts, and structured dialogue with mobile network operators and device importers. It is not a theoretical exercise. Every design choice described in the pages that follow reflects lessons learned from jurisdictions that have already travelled this road — some successfully, some less so — and from the documented failure modes of systems that were deployed without adequate attention to synchronization latency, privacy safeguards, or the political economy of device markets.

The problem is straightforward to state and difficult to solve: a mobile device's identity can be altered, duplicated, or fabricated, and the network has no inherent reason to notice. The IMEI, for all its limitations, remains the only globally standardized, manufacturer-assigned device identifier that every 3GPP-compliant network element is already designed to carry. The question is not whether to use it, but how to build an institutional and technical apparatus around it that is fast enough to matter, fair enough to survive legal challenge, and resilient enough to function when links fail and databases diverge.

What follows is a proposal to build that apparatus. It is deliberately detailed, because the failure of device identity systems rarely stems from a shortage of ambition. It stems from underspecified interfaces, unexamined data protection obligations, and the quiet assumption that a central database, once populated, will somehow enforce itself. This document resists that assumption at every turn.

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1. Executive Summary

1.1 The Problem in Brief

The International Mobile Equipment Identity (IMEI) is a fifteen-digit identifier assigned to every 3GPP-compliant mobile device at manufacture. It is intended to be unique, persistent, and tamper-resistant. In practice, it is frequently none of these things. Stolen devices continue to function on networks that do not enforce shared blacklists. Counterfeit devices carry fabricated or duplicated IMEIs that pass unnoticed because no authoritative reference exists against which to check them. Cloned IMEIs allow multiple physical devices to present the same identity to the network, undermining fraud detection, warranty enforcement, and law enforcement attribution.

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The consequences are not abstract. In Pakistan, the Device Identification, Registration and Blocking System (DIRBS) blocked nearly **100 million devices** in fiscal year 2024–25, including approximately **72 million counterfeit or replica handsets**, **27 million devices with duplicate or cloned IMEIs**, and **868,000 stolen or lost devices**. India's Central Equipment Identity Register, delivered through the Sanchar Saathi portal, has blocked over **38.5 lakh handsets** and recovered more than **6.07 lakh devices** since launch. These are not marginal interventions. They represent the difference between a device market that is governable and one that is not.

1.2 The Proposed Intervention

This proposal recommends the design, development, and phased deployment of a **National Central Equipment Identity Register (CEIR)** integrated with **operator-side Equipment Identity Registers (EIRs)** , following the **GSMA–CEIR–EIR architecture** validated across multiple jurisdictions and codified in **ITU-T Recommendation Q.5055**. The system will:

1. Maintain authoritative national white, grey, and black lists synchronized in near-real time with operator EIRs;
1. Validate every observed IMEI against the GSMA global TAC database and detect non-standard or counterfeit allocations;
1. Detect IMEI cloning through time-and-distance conflict analysis of call detail records and network attach events;
1. Provide a warrant-gated law enforcement module for device location and identification, with immutable audit logging;
1. Operate across 2G, 3G, 4G, and 5G network generations using SS7, Diameter (S13), and HTTP/2 service-based interfaces (N5g-EIR).

1.3 Why This Architecture, and Why Now

A purely centralized CEIR, while attractive on paper, introduces latency, single-point-of-failure risk, and operational fragility in environments where backhaul and power availability are uneven. A purely operator-side EIR model, by contrast, fragments enforcement: a device blocked on one network remains functional on another. The **hybrid architecture** — a central CEIR holding authoritative policy, with distributed EIR gateways enforcing locally — balances regulatory control with technical resilience. Nigeria's telecommunications regulator has explicitly endorsed this hybrid model, citing the successful precedents of India, Pakistan, and Turkey.

The timing is driven by three converging factors. First, the 5G service-based architecture has standardized the EIR interface (N5g-EIR) in **3GPP TS 29.511**, making multi-generational EIR deployment technically coherent for the first time. Second, GSMA has updated its TAC allocation rules under **TS.06 V27/V28 and TS.30 V25/V27**, including provisions for satellite bands and new equipment types, requiring national registries to refresh their validation logic. Third, the proliferation of eSIM and integrated SIM (iSIM) devices is beginning to erode the traditional separation between device identity and subscription identity, creating both new opportunities for authentication and new risks of identity obfuscation.

1.4 Indicative Investment and Timeline

The total indicative programme cost is in the range of **USD 4.0–6.5 million** over a **12-month phased implementation**, depending on the number of operators, existing core network readiness, and the chosen procurement model (build, buy, or managed service). This figure includes CEIR development and licensing, EIR deployment at operator sites, GSMA database integration, the law enforcement module, redundancy infrastructure, and legal/privacy assessment.

1.5 Recommendation

The Office recommends that the Regulatory Authority approve **Phase 1 (Design and Regulatory Alignment)** immediately, authorizing the commencement of a Data Protection Impact Assessment, formal stakeholder consultation, and the specification of CEIR–EIR synchronization protocols. Full system build and operator rollout should proceed only upon satisfactory completion of Phase 1 deliverables and the execution of binding data-sharing and enforcement agreements with participating operators.

2. Introduction and Background

2.1 Purpose of This Document

This document constitutes a full project proposal for the establishment of a national IMEI-based device tracking and management system. It is intended to serve simultaneously as a technical specification outline, a regulatory justification, an implementation roadmap, and a risk disclosure. It assumes a readership comprising telecommunications regulators, mobile network operators, device manufacturers and importers, law enforcement agencies, and data protection authorities. Where technical detail is included, it is included because the governance of such a system depends on an informed understanding of its mechanics; where legal and institutional considerations are raised, they are raised because a technically elegant system that cannot withstand legal scrutiny is not a system at all.

2.2 The IMEI as a Device Identity Primitive

The IMEI is a fifteen-digit decimal number structured in four principal components. The first eight digits constitute the **Type Allocation Code (TAC)** , which identifies the device model, brand owner, and original equipment manufacturer. The TAC is allocated exclusively by the GSMA through appointed Reporting Bodies — currently TÜV SÜD BABT, TAF, CTIA, and TIA — and only one TAC may be allocated to a single device model. The next six digits form the **Serial Number (SNR)** , assigned incrementally by the manufacturer. The final digit is a **Luhn check digit**, computed over the preceding fourteen digits, which allows immediate detection of transcription errors or trivial fabrication. An optional two-digit **Software Version Number (SVN)** may be appended, producing the sixteen-digit IMEISV.

The TAC is the linchpin of any validation system. Because every genuine device model carries a GSMA-allocated TAC, an observed IMEI whose TAC does not appear in the GSMA database is either misprogrammed or counterfeit. Because one TAC supports a maximum of **one million devices**, a TAC observed on more than one million distinct IMEIs is either being reused illegitimately or indicates a programming error at the manufacturing stage. These two simple checks, applied at network attach, eliminate a substantial proportion of counterfeit and misprogrammed devices without any need for device-side cooperation.

2.3 Why IMEI Tracking Is Necessary but Insufficient

It is important to state plainly what an IMEI tracking system can and cannot do. It **can** deny service to a device whose IMEI appears on an authoritative blacklist. It **can** detect that two devices are presenting the same IMEI simultaneously and flag the anomaly. It **can** provide law enforcement with a standardized identifier that correlates to a physical handset, independent of the SIM card currently inserted. It **cannot**, by itself, determine who is using a device, where a device is located with precision beyond cell-level granularity (absent additional location infrastructure), or whether a device was obtained lawfully by its current possessor. It is an **equipment identity** system, not a subscriber identity system, and the distinction matters for privacy, for legal process, and for the design of proportionate enforcement rules.

The system proposed here is therefore deliberately scoped. It does not replace lawful interception, subscriber registration, or international mutual legal assistance. It provides a **device-layer control plane** that complements those functions and fills a gap that subscriber-centric systems leave open: the ability to act on the device itself, irrespective of the SIM inside it.

2.4 Lessons from Comparative Implementations

Pakistan offers the most extensive operational evidence base. DIRBS, developed in collaboration with Qualcomm and launched as an open-source platform, combines device registration at import, network-based IMEI observation, and a centralized blocking system. Its enforcement statistics — nearly 100 million devices blocked in a single fiscal year — demonstrate both the scale of the counterfeit problem and the operational feasibility of national-scale IMEI control. The system's success is attributable in part to its integration with customs and import verification, which prevents unregistered devices from entering the market in the first place.

India's CEIR, delivered through the citizen-facing Sanchar Saathi portal, has demonstrated the value of a **self-service interface** for device blocking. Users can report a lost or stolen handset and have it blocked across all networks through a single online form, with law enforcement able to trace location if the device is subsequently used. The Indian experience also illustrates a governance lesson: the system's effectiveness depends on the speed with which blocklist updates propagate to operator EIRs, and on the legal clarity of the process by which a device is **unblocked** when recovered or when a blocking order is reversed.

Turkey maintains a long-standing IMEI registration regime with a national database covering hundreds of millions of device records. Its experience with cloned IMEIs — at various points estimated in the millions — underscores the importance of **pairing analysis**, not merely list matching. A cloned IMEI will not appear on any blacklist until it is detected as a duplicate, which requires active comparison of IMEI–IMSI pairs across time and geography.

Nigeria is currently implementing a Device Management System under a public-private partnership, with the regulator explicitly adopting the hybrid CEIR-plus-distributed-EIR model and citing India, Pakistan, and Turkey as reference cases. The Nigerian design notes emphasize **fail-open behavior at the EIR level** (so that a CEIR link outage does not strand every subscriber) and **local enforcement capability** (so that blocking decisions survive central connectivity interruptions) — both of which are incorporated into the architecture proposed here.

Colombia provides a useful model for the **procedural handling of duplicate IMEIs**. When a duplicated IMEI is detected, users are given thirty calendar days to present purchase documentation. The operator then determines which user has the genuine device based on registration and documentary evidence, associates the correct IMEI–IMSI pairing in the EIR, and blocks the duplicates from the centralized negative database. This graduated approach — detection, notification, evidentiary process, enforcement — is more legally durable than immediate blanket blocking and is recommended for adaptation in the present proposal.

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3. Technical Architecture

3.1 Architectural Overview

The system adopts a **five-layer distributed architecture** consistent with **ITU-T Q.5055** and the GSMA CEIR framework. Each layer performs a distinct function and is coupled to adjacent layers through standardized interfaces. The layers are:

Layer	Entity	Core Function	Primary Data
1. Manufacturer	Device OEMs	IMEI generation and TAC programming	TAC, SNR
2. Global Registry	GSMA	Global TAC allocation and IMEI database	GSMA IMEI DB
3. National Registry	**CEIR**	Authoritative national device control; list synchronization; inter-operator exchange	White/Grey/Black lists
4. Operator Registry	**EIR**	Local IMEI verification at network attach; policy enforcement	Local synchronized lists
5. Network Core	MNO/MVNO	Authentication and access decision	HSS/UDM, AMF, MME, SGSN

The design principle is **centralized policy, distributed enforcement**. The CEIR holds the authoritative lists and the validation logic. The EIRs hold synchronized local copies and apply the logic at the point of network attach. This division allows the CEIR to remain the single source of truth while ensuring that enforcement decisions do not depend on continuous, low-latency connectivity to the centre.

3.2 The CEIR: Functions and Responsibilities

The CEIR is the institutional and technical heart of the system. Its responsibilities include:

1. **Maintaining the national device register**, comprising all IMEIs observed on national networks, their first-seen and last-seen timestamps, associated TACs, and status history.
2. **Synchronizing with the GSMA IMEI Database** to validate TACs, detect non-standard allocations, and receive the GSMA global blacklist where available under the **SG.18 file format** provisions.
3. **Managing the white, grey, and black lists** , including additions, removals, expiry, and override mechanisms. A **Blacklist Override (BLO)** function allows specific IMEI–IMSI pairs to be permitted even when the IMEI is otherwise blocked, addressing the legitimate-user exception problem in cloning cases.
4. **Broadcasting list changes to operator EIRs** using a synchronous API-based `CairBroadcast` message, with acknowledgement handling and link-recovery resynchronization.

5 **Detecting duplicate IMEIs** through national-level cross-operator analysis of call detail records and attach events, using collision and time/distance conflict algorithms.

6 **Providing a law enforcement interface** with warrant verification, case binding, and immutable audit logging.

7 **Generating operational and strategic reports** for the regulator, including enforcement statistics, cloning hotspots, and TAC anomaly trends.

The CEIR should be deployed in a **high-availability active-active configuration** with geo-redundant database replication. The indicative capacity specification is ≥**100 million individual IMEI records** and ≥**10 million TAC/range records**, with headroom for five years of growth.

3.3 The EIR: Functions and Deployment Models

The EIR is the enforcement point. It resides within or adjacent to the operator's core network and is queried during device attach. Its functions are deliberately minimal to maximize performance and resilience:

1 Receive and store synchronized copies of the CEIR white/grey/black lists;

1 Answer IMEI check requests from core network elements with a verdict of **WHITELISTED**, **GREYLISTED**, or **BLACKLISTED**;

1 Apply configurable matching logic (exact IMEI, IMEI range, IMEISV, IMSI);

1 Fail open or fail closed according to regulatory policy, with fail-open as the recommended default to prevent mass service denial during CEIR link outages;

1 Log all queries for audit and synchronization reconciliation.

EIRs may be deployed as **standalone network functions**, **integrated into the HSS/UDM**, or **provided as a managed service** by a third-party vendor. The choice should be left to each operator, subject to interoperability certification against the CEIR interface specifications.

3.4 Multi-Generational Interface Support

A critical design requirement is that the EIR must serve devices across 2G, 3G, 4G, and 5G networks. The interface landscape is as follows:

2G/3G (SS7/MAP): The EIR is queried via the **MAP Check IMEI** procedure. The SGSN or MSC sends the IMEI to the EIR and receives a status response. This interface remains relevant for legacy devices and for networks that have not yet completed 4G/5G migration.

4G (Diameter S13): The MME queries the EIR over the **S13 interface** using the Diameter `ME-Identity-Check` request/answer pair. The EIR identifies the mobile equipment based on the first fourteen digits of the IMEI AVP, with the fifteenth digit serving as the check digit and the optional SVN appended where present. The S13 interface is specified in **3GPP TS 29.272** and is widely implemented in commercial EIR products.

5G (HTTP/2 Service-Based Interface — N5g-EIR): The AMF invokes the `N5g-EIR_EquipmentIdentityCheck` service, a RESTful `GET` operation over HTTP/2, specified in **3GPP TS 29.511**. The request carries the **Permanent Equipment Identifier (PEI)** — formatted as `imei-<15 digits>` or `imeisv-<16 digits>` — and optionally the **SUPI** (subscription identifier). The response is an `EirResponseData` object containing an `EquipmentStatus` of `WHITELISTED`, `GREYLISTED`, `BLACKLISTED`, `NOT_VERIFIED`, or `REJECTED`, with optional problem details and cause codes. The interface mandates **mutual TLS (mTLS)** , OAuth 2.0 client credentials, and compliance with **3GPP TS 33.501** privacy requirements, including mandatory masking of IMEI in logs.

The N5g-EIR specification also introduces a **verdict caching mechanism** (via `Cache-Control` and `ETag`) allowing the AMF to cache short-lived results and reduce EIR load, and an **asynchronous notification subscription** (`N5g-eir_Notify`) enabling the EIR to push blocklist changes to registered network functions — a significant advance over the polling or broadcast-only models of earlier generations.

A well-designed EIR implementation should expose **a unified rule set across S13 and N5g-EIR**, so that a device blacklisted on 4G is automatically blacklisted on 5G with no duplicate provisioning. Commercial implementations already demonstrate this capability, matching equipment checks in the order **IMEISV → IMEI → IMSI, first regex match wins**.

3.5 IMEI Cloning Detection Architecture

Cloning detection is not a list-matching problem; it is an **anomaly detection** problem. The system must identify when the same IMEI is observed in association with different IMSIs in a manner that is physically implausible or statistically anomalous.

The primary detection method is **time/distance conflict analysis**. For each IMEI observed on the network, the system records the serving cell identity, timestamp, and associated IMSI. If the same IMEI is observed on two cells separated by a distance that could not be traversed in the elapsed time — accounting for the maximum plausible speed of the device — a cloning alert is generated. This analysis can be performed **daily at the operator level** using call detail records, and **monthly at the national level** using aggregated data across operators. The national-level analysis is essential because a clone may operate on a different network from the genuine device, and only a cross-operator view will reveal the conflict.

Secondary detection methods include:

1. **IMEI–IMSI pairing stability analysis:** A genuine device typically retains a stable or slowly changing set of IMSI associations. A device whose IMEI is associated with a rapidly rotating set of IMSIs, or with IMSIs from a different subscriber segment, may be a clone or a fraud device.
1. **TAC–RAT consistency checks:** The TAC encodes the device model's radio access technology capabilities. A 2G-only TAC observed on a 4G or 5G network is an indicator of misprogramming or fraud.
1. **eUICC authentication (where available):** For eSIM/iSIM devices, the embedded UICC can provide a cryptographic binding between the device and the subscription that is not present in removable-SIM devices. Where such authentication is available, it can be used to distinguish the genuine device from a clone in a duplicate-IMEI scenario.

Upon detection, the system should follow a **graduated response**: alert the operator and the CEIR, initiate the evidentiary process described in Section 3.6, and apply a **temporary greylist status** to the IMEI pending resolution, rather than immediate blacklisting of all devices presenting that IMEI. This protects the genuine user while containing the clone.

3.6 The Duplicate IMEI Resolution Workflow

Drawing directly from the Colombian model, the system shall implement the following workflow:

1. **Detection:** National or operator-level analysis identifies a duplicated IMEI and the set of IMSIs associated with it.
2. **Notification:** All affected users are notified through their operator and given **thirty calendar days** to present purchase documentation, point-of-sale information, and any other evidence of lawful acquisition.
3. **Evidentiary assessment:** The operator assesses the documentation against its registration records and any import/customs data available through the CEIR.
4. **Determination:** The genuine IMEI–IMSI pairing (or pairings, in the case of legitimate device transfers) is added to the **Blacklist Override (BLO)** exceptions list. The IMEI is placed on the blacklist for all other pairings.
5. **Enforcement:** The clone is blocked; the genuine device continues to receive service.
6. **Escalation:** Where evidence of IMEI tampering or trafficking is found, the matter is referred to law enforcement.

This workflow is more administratively burdensome than blanket blocking, but it is **legally sustainable** and avoids the severe user harm that indiscriminate blocking would cause. The CEIR must support the workflow natively, including document upload, case tracking, and time-bound status transitions.

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4. Functional Requirements

4.1 List Management

The system shall support three principal list categories with the following semantics:

Whitelist: IMEIs verified as genuine and permitted full network access. The whitelist may be **explicit** (device registered at import or through a citizen self-registration portal) or **implicit** (TAC validated against GSMA, no adverse record). The system should support both modes and allow the regulator to choose the default posture.

Greylist: IMEIs under observation or subject to a pending determination. Greylisted devices are permitted to attach but are **tracked with enhanced logging**, and may be subject to periodic review or automatic escalation to blacklist if the triggering condition persists. Greylist status is the appropriate initial state for duplicate-IMEI detection and for devices whose TAC is valid but whose provenance is uncertain.

Blacklist: IMEIs denied service. Blacklisting grounds include confirmed theft, confirmed fraud, duplicate IMEI (for all but the determined genuine pairing), and TAC invalidity where the device is found to be counterfeit. The system must support **scheduled blacklisting** (e.g., at the expiry of a grace period) and **immediate blacklisting** (for stolen devices reported by law enforcement).

Blacklist Override (BLO): A distinct exception mechanism that permits a specific IMEI–IMSI pair to receive service notwithstanding the IMEI's blacklisted status. BLO entries are time-bound, attributable to an authorized officer, and subject to audit. They are essential for handling legitimate users caught in cloning events and for lawful exceptions granted by the regulator.

4.2 TAC Validation and Counterfeit Detection

Every IMEI observed on the network shall be subjected to the following validation sequence:

1. **Length and format check:** Fifteen digits (or sixteen for IMEISV), all numeric.
2. **Luhn check digit validation:** Reject IMEIs that fail the Luhn algorithm.
3. **TAC existence check:** Query the GSMA IMEI Database (or a synchronized local cache) to confirm that the first eight digits correspond to an allocated TAC.
4. **TAC allocation status check:** Confirm that the TAC has not been withdrawn, suspended, or flagged for security concerns.
5. **TAC capacity sanity check:** Flag TACs observed on an implausible number of distinct devices relative to the one-million-unit allocation limit.
6. **TAC–RAT consistency check:** Compare the device capabilities implied by the TAC with the network generation on which the device is observed.

IMEIs failing steps 1–4 are candidates for **immediate blacklisting** as counterfeit or misprogrammed. Failures at steps 5–6 warrant **greylisting and investigation**, as they may arise from legitimate manufacturing anomalies or from cloning.

4.3 Real-Time Synchronization and Broadcast

The CEIR–EIR synchronization interface shall support two modes:

Synchronous API-based broadcast: When a list change occurs at the CEIR, a `CairBroadcast` message is sent to all connected EIRs. Each EIR acknowledges with `CairBroadcastResp`, indicating successful addition or deletion of the entry. If an EIR is unreachable, the CEIR queues the update and delivers it upon link recovery, including a **link-down data reconciliation** payload.

Asynchronous API-based query and broadcast: For high-load conditions or unreliable links, the EIR may temporarily allow a device with an **unverified IMEI–IMSI pair** while placing that pair on a local **BLO pending** list, and query the CEIR asynchronously. The CEIR responds via broadcast, and the EIR reconciles its local state. This mode prevents the CEIR from becoming a real-time bottleneck during network congestion or synchronization storms.

The system shall expose synchronization health metrics, including **lag time per EIR**, **broadcast success rate**, and **queue depth**, and shall raise alarms when lag exceeds a configurable threshold (recommended: 5 minutes for blacklist changes, 30 minutes for whitelist changes).

4.4 Law Enforcement Module

The law enforcement module is the most sensitive component of the system. It shall be designed on the principle of **warrant-gated access with immutable audit**. Its functional requirements are:

1. **Warrant registration:** Every tracking request must be associated with a judicial warrant identifier, the issuing court, the case reference, and the requesting officer's credentials. The system shall validate the warrant against a pre-registered warrant registry (either integrated or via a secure API to the judiciary).

1. **Tracking request types:** (a) **IMEI status query** — is a given IMEI currently attached to the network, and on which cell? (b) **IMEI–IMSI association query** — which IMSIs have been observed with a given IMEI over a specified period? (c) **Cell-level location** — the serving cell and timing advance data for an active device; (d) **Historical mobility reconstruction** — the sequence of cells visited by a device over a period, derived from attach and handover records.
1. **No direct GPS access:** The system does not itself trigger GPS pinging. Where GPS data is available through emergency location infrastructure (e.g., AML/ELS), access must be through a separate, independently authorized channel. The CEIR's law enforcement module provides **network-derived location**, which is cell-level or better depending on network density.
1. **Immutable audit log:** Every query, every result, and every export is logged with a cryptographic hash chain. Logs are write-once and retained for a minimum period specified by the data protection authority (recommended: five years for law enforcement access logs, with automated deletion thereafter unless flagged for an ongoing investigation).
1. **Supervisory approval:** High-sensitivity queries (historical mobility reconstruction, bulk IMSI association) require a second-level approval within the law enforcement agency, recorded in the system.
1. **Breach notification:** Any unauthorized access attempt or anomalous query pattern triggers immediate notification to the regulator's security officer and the data protection authority.

4.5 Reporting and Analytics

The system shall provide:

1. **Operational dashboards:** real-time attach volumes, check verdict distributions, list sizes, synchronization lag, and EIR availability.
1. **Enforcement analytics:** blacklist hit rates by device model, TAC, and geographic region; duplicate-IMEI detection trends; counterfeit TAC clusters.
1. **Strategic reports:** quarterly device market composition (genuine/counterfeit/grey), import-channel analysis where customs data is integrated, and law enforcement outcome tracking (devices recovered, arrests facilitated, cases supported).
1. **Configurable alarms:** abnormal attach failure rates, sudden spikes in blacklist hits (potential coordinated fraud), synchronization failures, and unusual law enforcement query patterns.

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5. Legal, Privacy, and Regulatory Framework

5.1 The Legal Character of the IMEI

Under the **General Data Protection Regulation (GDPR)** and analogous data protection regimes, the IMEI is **personal data**. The European Parliament's legislative analysis explicitly identifies the IMEI, alongside the MAC address and IMSI, as a unique identifier emitted by mobile devices that enables singling out and tracking of individuals. The Spanish data protection authority (AEPD) has similarly concluded that the IMEI and IMSI, insofar as they permit the singling out of an individual, must be treated as personal data. This characterization has direct consequences for the system's design: purpose limitation, data minimization, storage limitation, integrity and confidentiality, and accountability must all be operationalized, not merely asserted.

5.2 Purpose Limitation and Legal Basis

The system shall process IMEI data only for the following purposes, each of which must have a clear legal basis in national law:

1. **Device authentication and network integrity** (legitimate interest / regulatory mandate);
2. **Theft prevention and device recovery** (legal obligation / public task);
3. **Counterfeit suppression** (public task / consumer protection);
4. **Law enforcement support under judicial warrant** (legal obligation / administration of justice).

Any secondary use — including marketing analytics, commercial device-market research, or cross-border data sharing beyond the scope of a formal agreement — is prohibited. The CEIR shall maintain a **purpose registry** documenting the legal basis for each processing activity, reviewed annually by the data protection authority.

5.3 Data Minimization and Retention

The system shall collect only the data necessary for each purpose. Specifically:

1. **IMEI and TAC** are retained for the life of the device registration plus the statutory audit period.
1. **IMEI–IMSI association records** are retained for a period sufficient to support cloning detection and law enforcement investigations (recommended: **12 months**, with extension only under a warrant).
1. **Network attach logs** (cell, timestamp, verdict) are retained for **90 days** in identifiable form, and thereafter in aggregated or anonymized form for analytics.
1. **Law enforcement query logs** are retained for **five years**, with automated deletion thereafter unless preserved under a litigation hold.
1. **Tracking results** (location, mobility history) are retained only for the duration of the authorized investigation and are destroyed upon case closure, subject to judicial retention orders.

Automated deletion jobs shall be independently audited, and deletion certificates shall be generated and retained.

5.4 Access Control and Audit

Access to the CEIR shall be governed by **role-based access control (RBAC)** with the following roles: Regulator Administrator, Regulator Analyst, Operator Liaison, Law Enforcement Officer (warrant-gated), Law Enforcement Supervisor (approval authority), and Data Protection Auditor (read-only audit access). All access shall require **multi-factor authentication** and shall be logged with user identity, timestamp, action, and purpose code.

The audit log shall be **tamper-evident** through a hash chain or equivalent mechanism, and shall be accessible to the data protection authority without prior approval from the system operator.

5.5 Cross-Border Data Sharing

IMEI data may be shared with foreign CEIRs or with the GSMA global database only under a **formal bilateral or multilateral agreement** that specifies the categories of data shared, the purposes, the retention period, the security measures, and the redress mechanism for data subjects. The system shall maintain a **data sharing register** documenting all such agreements, and shall support **selective sharing** (e.g., sharing blacklist entries without sharing associated IMSI or subscriber data).

5.6 Regulatory Compliance with GSMA and ITU Instruments

The system shall be designed to comply with:

1. **GSMA TS.06** (IMEI Allocation and Approval Process) and **TS.30** (TAC Application Forms);
1. **GSMA SG.18** (file format for blacklist exchange);
1. **ITU-T Q.5050, Q.5051, Q.5052, and Q.5055** (CEIR technical requirements, interfaces, and functions);
1. **3GPP TS 29.272** (S13 Diameter interface), **TS 29.511** (N5g-EIR), and **TS 33.501** (5G security and privacy);
1. Applicable national data protection and electronic communications legislation.

Where conflicts arise between international specifications and national law, national law prevails, and the system shall be configured to meet the stricter of the two standards.

6. Implementation Plan

6.1 Phased Approach

The implementation is structured in five phases over approximately twelve months, with a decision gate at the end of Phase 1.

Phase 1: Design, Legal Alignment, and Stakeholder Formalization (Months 1–2)

Deliverables: Data Protection Impact Assessment (DPIA); legal basis memorandum; stakeholder agreements (regulator, operators, law enforcement, customs); CEIR–EIR interface specification; procurement strategy; project governance charter.

Decision gate: Approval of the DPIA by the data protection authority; execution of binding participation agreements by operators representing at least 80% of national mobile connections.

Phase 2: CEIR Development and GSMA Integration (Months 3–6)

Deliverables: CEIR core platform (list management, TAC validation, broadcast engine, duplicate detection engine); GSMA IMEI Database integration; law enforcement module (warrant verification, audit logging); reporting and analytics dashboard; high-availability infrastructure.

Acceptance criteria: Successful TAC validation against the GSMA database for a test set of 10,000 genuine and 1,000 counterfeit IMEIs; broadcast synchronization to a simulated EIR within 60 seconds; audit log integrity verification.

Phase 3: Operator EIR Deployment and Integration (Months 7–9)

Deliverables: EIR deployment at each participating operator (standalone, integrated, or managed service); S13 and N5g-EIR integration testing; synchronization link provisioning and redundancy; fail-open/fail-closed configuration validation; end-to-end attach-flow testing for 2G/3G/4G/5G devices.

Acceptance criteria: A blacklisted IMEI is denied service on all participating networks within 5 minutes of CEIR entry; a whitelisted device attaches successfully across all generations; a simulated CEIR outage does not disrupt service for non-blacklisted devices.

Phase 4: Law Enforcement Training, Privacy Audit, and Pilot (Months 10–11)

Deliverables: Law enforcement training programme (warrant process, query types, audit obligations); independent privacy audit; pilot operation with two to three operators covering at least 30% of national connections; pilot evaluation report.

Acceptance criteria: Pilot demonstrates detection of at least one real or simulated clone; law enforcement queries are successfully warrant-gated and audited; privacy audit identifies no material non-compliance.

Phase 5: National Rollout and Stabilization (Month 12 and ongoing)

Deliverables: Full operator onboarding; citizen-facing device verification and blocking portal (where included in scope); public communication campaign; operational handover to the regulator’s permanent device identity unit; first quarterly enforcement report.

6.2 Procurement Considerations

The Authority may choose among three procurement models:

Build (in-house or contracted development): Maximum control and customization; highest upfront cost and longest timeline; requires sustained in-house technical capacity for maintenance.

Buy (commercial CEIR/EIR platform): Faster deployment; proven feature set; vendor dependence for customization and roadmap; licensing costs scale with subscriber volume.

Managed service (PPP or concession): Lowest upfront capital expenditure; risk transfer to the operator; requires robust contract design to ensure data ownership, audit rights, and exit provisions. Nigeria has adopted a PPP model for its DMS, and the Authority may wish to study the Nigerian contract structure as a reference.

A **hybrid approach** — commercial CEIR platform with in-house integration and customization — is likely to offer the best balance of speed, cost, and control for a first-generation deployment.

6.3 Capacity Building and Change Management

The system will not function without trained people. The programme shall include:

1. **Regulator staff:** CEIR administration, list management, dispute resolution, and reporting (minimum 8–10 trained staff).
1. **Operator staff:** EIR operations, synchronization troubleshooting, and escalation procedures (minimum 3–5 trained staff per operator).
1. **Law enforcement:** warrant process, query procedures, and audit obligations (minimum 20–30 trained officers across relevant units).
1. **Customs and import authorities:** device registration and TAC verification at border (where customs integration is in scope).

Training shall be competency-based, with certification and refresher requirements.

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7. Stakeholder Analysis and Governance

7.1 Stakeholder Roles

Stakeholder	Primary Interest	Role in the System	Risk if Disengaged
Telecommunications Regulator	Market integrity, consumer protection, legal compliance	CEIR owner and operator; policy-setting; enforcement oversight	System lacks authoritative governance; legal challenges
Mobile Network Operators	Network security, operational cost, subscriber satisfaction	EIR deployment and operation; enforcement at network edge	Incomplete enforcement; device blocking is ineffective
Device Manufacturers and Importers	Market access, compliance cost	GSMA TAC registration; device registration at import	Counterfeit devices enter supply chain; legitimate devices incorrectly blocked
Law Enforcement	Investigative capability, legal defensibility	Warrant-gated use of tracking module	Module unused or misused; evidence challenged in court
Data Protection Authority	Privacy compliance, data subject rights	DPIA review; audit access; complaint handling	Legal challenge; reputational damage; suspension orders
Judiciary	Due process, proportionality	Warrant issuance and oversight	Tracking without lawful authority; evidence inadmissible
GSMA	Global standards coherence	TAC allocation; IMEI database access	National system diverges from global standards; interoperability loss

7.2 Governance Structure

The programme shall be governed by a **Steering Committee** chaired by the Regulator and comprising representatives of the operators, the law enforcement liaison office, the data protection authority, and the customs authority. The Steering Committee shall meet monthly during implementation and quarterly thereafter. A **Technical Working Group** shall handle interface specifications, synchronization protocols, and incident response. An **Independent Audit Committee** , with at least one member appointed by the data protection authority, shall review audit logs and privacy compliance annually.

7.3 Dispute Resolution

Disputes between stakeholders — for example, an operator contesting a blacklisting decision, or a subscriber challenging a device block — shall be resolved through a defined escalation path: operational review (CEIR administrator) → regulatory review (Regulator) → independent appeal (designated ombudsman or tribunal). The system shall support **case management** for disputes, including document upload, status tracking, and outcome recording.

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8. Risk Assessment and Mitigation

The following risk register identifies the principal risks to successful delivery and operation, with likelihood, impact, and mitigation measures. This register should be treated as a living document, reviewed at each Steering Committee meeting.

Risk ID	Risk Description	Likelihood	Impact	Mitigation
R1	**IMEI spoofing or reprogramming** defeats list-based enforcement	High	High	Combine list matching with TAC validation, IMEI–IMSI pairing analysis, and time/distance conflict detection; treat list matching as necessary but not sufficient
R2	**International blacklist evasion:** stolen devices trafficked across borders and reactivated	High	Medium	Pursue bilateral CEIR data-sharing agreements; engage GSMA for global blacklist propagation; integrate border customs registration
R3	**Privacy litigation or regulatory suspension**	Medium	High	Complete DPIA before build; implement privacy-by-design; ensure data protection authority has audit access and a formal role in governance
R4	**Operator non-compliance** with EIR deployment or list synchronization	Medium	Medium	Make EIR deployment a licence condition; implement tiered enforcement (warning, fine, licence review); provide shared EIR service for small operators
R5	**CEIR–EIR synchronization failure** leaves stale lists and inconsistent enforcement	Medium	High	Implement queued broadcast with link-recovery reconciliation; monitor lag per EIR; fail-open at EIR with local BLO pending list
R6	**False positive blocking** of legitimate devices (e.g., due to TAC data errors or cloning collateral)	Medium	High	Implement BLO exceptions; graduated response with grace period; accessible dispute resolution; periodic TAC data quality audits
R7	**Law enforcement module misuse** or unauthorized access	Low	High	Warrant-gated access; immutable audit; supervisory approval for sensitive queries; independent audit; breach notification
R8	**Scalability bottleneck** as IMEI volume and query rate grow	Low	Medium	Deploy horizontally scalable CEIR architecture; load-test with simulated volumes; use verdict caching at EIR (N5g-EIR Cache-Control)
R9	**Vendor lock-in** limits future customization or increases costs	Medium	Medium	Require open APIs and documented data models; retain ownership of data; include exit and data portability clauses in contracts
R10	**Political or industry resistance** delays implementation	Medium	Medium	Early stakeholder engagement; transparent governance; pilot demonstration of benefits; phased approach with clear decision gates

9. Budget Estimate

9.1 Capital and Operating Cost Categories

The following table provides an indicative budget for a national deployment covering approximately ten mobile network operators and a CEIR serving a subscriber base in the range of 50–100 million connections. Costs are expressed in United States dollars and are indicative only; final costs will depend on procurement model, vendor selection, and scope decisions.

Category	Description	Indicative Cost (USD)
CEIR platform	Software licensing, customization, integration, and first-year support	$1,200,000 – $2,000,000
EIR deployment (per operator)	Standalone EIR hardware/software or managed service setup, integrated with core network	$150,000 – $250,000 per operator; **$1,500,000 – $2,500,000 total for 10 operators**
GSMA integration	IMEI Database access, TAC validation API, SG.18 blacklist exchange	$150,000 – $250,000
Law enforcement module	Warrant verification integration, audit logging, secure access infrastructure	$300,000 – $500,000
Infrastructure and redundancy	Cloud or on-premise compute, storage, geo-redundant replication, backup	$400,000 – $700,000
Professional services	DPIA, legal drafting, project management, systems integration	$250,000 – $400,000
Training and change management	Curriculum development, delivery, certification, materials	$150,000 – $250,000
Independent audit and assurance	Pre-launch privacy audit, security penetration testing, operational audit	$80,000 – $150,000
Contingency (10–15%)		$400,000 – $900,000
Total indicative		**$4,430,000 – $7,650,000**

9.2 Funding Options

The Authority may consider:

1. **Direct government appropriation** (capital budget), with operating costs recovered through regulatory fees or operator levies;
1. **Public-private partnership** (as in Nigeria), with a concessionaire bearing deployment costs in exchange for a revenue share or service fee;
1. **Operator-funded model** with regulatory mandate, where operators bear EIR costs and the CEIR is funded through a universal service or regulatory fee;
1. **Development partner or multilateral funding** where the system is positioned as a consumer protection and market integrity intervention.

A detailed financial model, including cash-flow projections, cost recovery scenarios, and sensitivity analysis, should be developed during Phase 1.

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10. Monitoring, Evaluation, and Learning

10.1 Performance Indicators

The programme shall be evaluated against the following key performance indicators:

Indicator	Baseline (pre-launch)	Target (12 months post-launch)	Target (36 months)
Blacklist synchronization lag (CEIR to EIR)	Not applicable	≤5 minutes (95th percentile)	≤2 minutes
Share of national connections subject to EIR enforcement	0%	≥80%	≥95%
Duplicate IMEI detection rate (devices flagged per 100,000 active IMEIs)	Unknown	Establish baseline	20% reduction in detected duplicates year-on-year
Counterfeit TAC detection rate	Unknown	Establish baseline	≥90% of observed counterfeit devices flagged
Law enforcement query response time	Not applicable	≤30 minutes for active-device cell query	≤10 minutes
Dispute resolution turnaround	Not applicable	≤30 calendar days	≤15 calendar days
Privacy audit findings	Not applicable	No material findings	No material findings; all recommendations closed within 90 days

10.2 Evaluation Methodology

A **baseline study** shall be conducted before launch to establish the prevalence of counterfeit, cloned, and stolen devices in the national market, using a combination of operator CDR analysis, customs data, and market surveys. A **mid-term evaluation** at 12 months shall assess operational performance, enforcement outcomes, and stakeholder satisfaction. A **full evaluation** at 36 months shall assess the system's impact on device theft, counterfeit prevalence, and law enforcement outcomes, and shall recommend adjustments to the list management rules, governance structure, or technical architecture.

10.3 Learning and Adaptation

The system shall be designed for **iterative improvement**. The rule set for greylisting, the grace period for duplicate IMEI resolution, the thresholds for cloning detection, and the list synchronization architecture should all be treated as **configurable policy parameters**, not hard-coded constants. A **change control process** shall allow the regulator to adjust parameters based on operational experience, with appropriate consultation and documentation. An annual **lessons-learned review** shall compare national experience with international developments, including GSMA specification updates and the experience of other CEIR operators.

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11. Conclusion and Recommendation

The proposal set out in these pages is ambitious but not novel. Every major element — the CEIR, the EIR, the GSMA TAC framework, the N5g-EIR interface, the graduated duplicate-IMEI workflow — has been implemented and tested in at least one jurisdiction. What is novel is the **combination**: a hybrid architecture that is fast enough to enforce in real time, resilient enough to survive link failures, transparent enough to withstand privacy scrutiny, and governed by a stakeholder structure that distributes authority without diluting accountability.

The case for proceeding is strong. Pakistan's experience demonstrates that national-scale IMEI control can block tens of millions of non-compliant devices in a single year. India's experience demonstrates that a citizen-facing blocking portal can recover hundreds of thousands of devices and create a tangible public benefit that sustains political support. Colombia's experience demonstrates that a graduated, evidence-based approach to duplicate IMEIs is legally and operationally feasible. The technical standards are mature. The privacy framework is demanding but navigable. The risks are real but manageable with the mitigation measures described in Section 8.

The Office therefore recommends that the Regulatory Authority:

1. **Approve Phase 1 immediately**, authorizing the commencement of the Data Protection Impact Assessment, stakeholder formalization, and interface specification.
2. **Establish the Steering Committee and Technical Working Group** within 30 days of approval, with named representatives from each stakeholder category.

3. **Commission a detailed financial model and procurement strategy** during Phase 1, to be presented alongside the DPIA for the Phase 1 decision gate.
4. **Commit to the phased, gated approach** described in Section 6, with no irreversible capital commitment until the Phase 1 gate is passed.
5. **Communicate the programme publicly** in terms of consumer protection, device recovery, and market integrity — not in terms of surveillance — and ensure that the law enforcement module is presented, accurately, as a warrant-gated investigative tool operating within the existing legal framework.

The problem of device identity is not going away. As devices proliferate, as eSIM and iSIM blur the line between equipment and subscription, and as cross-border trafficking of stolen and counterfeit handsets continues, the absence of a national IMEI control system will become an increasingly conspicuous gap in the regulatory architecture. The proposal before the Authority is an opportunity to close that gap with a system that is technically sound, legally defensible, and operationally proven.

Annexes (Indicative — to be developed in Phase 1)

1. **Annex A:** CEIR–EIR Interface Control Document (ICD) outline, covering synchronous and asynchronous API specifications.
1. **Annex B:** Data Protection Impact Assessment methodology and template.
1. **Annex C:** Stakeholder participation agreement template.
1. **Annex D:** Law enforcement warrant verification protocol.
1. **Annex E:** Duplicate IMEI resolution workflow — detailed process maps.
1. **Annex F:** Glossary of technical terms and abbreviations.
1. **Annex G:** Reference documents register (GSMA TS.06, TS.30, SG.18; ITU-T Q.5050–Q.5055; 3GPP TS 29.272, TS 29.511, TS 33.501).

Document sign-off

Role	Name	Signature	Date
Director, Office of Device Identity and Network Integrity			
Chief Technical Officer, Regulatory Authority			
Legal Counsel, Regulatory Authority			
Chair, Steering Committee (approval)			

This document contains forward-looking estimates and projections that are subject to the outcomes of Phase 1 and to procurement processes. Figures are indicative and do not constitute a financial commitment. Technical specifications are based on GSMA and 3GPP documentation available as of the date of preparation and are subject to revision as standards evolve.

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